

CHEN-YE HONG

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Research Interests

Brain–computer interfaces, with a focus on using neural and physiological signals to inform how interactive systems adapt to users’ intentions and states.

Education

National Taiwan University (NTU)

B.S. in Mechanical Engineering

GPA: 4.16/4.30 | Graduation Rank: 3/146

Sep. 2022 – Jan. 2026

Taipei, Taiwan

National Tsing Hua University (NTHU)

Undergraduate Studies in Power Mechanical Engineering

GPA: 4.16/4.30 | Department Rank: 1/116

Sep. 2021 – Jun. 2022 (Transferred to NTU)

Hsinchu, Taiwan

Publications

[†]Equal contribution (co-first authors).

- **C.-Y. Hong[†]**, C.-H. Chou[†], Y.-J. Wang, K.-L. Wu, S.-J. Yang, C.-H. Chang, and D.-R. Li (2026). “An experimental cadaveric limb motion platform for detecting altered mechanical responses under small-joint defect conditions.” *Medical Engineering & Physics*, 147(9), 095006. [DOI]
- C.-Y. Hsieh[†], **C.-Y. Hong[†]**, J.-Z. Li[†], P.-C. Yang[†], Y.-C. Hsu, C.-H. Tsai, C.-H. Hsu, and Y.-T. Wang (2026). “From Movement to Manipulation: Multimodal Control of a Mobile Assistive Robot.” *Proceedings of the 14th International Conference on Brain–Computer Interface (BCI)*, IEEE. (Oral presentation) [DOI]

Submitted Manuscripts

- **C.-Y. Hong**, I. C. Lee, C.-C. Deng, J. P. Hsu, and Y. T. Wang. “RehabVerse: A Virtual-Reality Multiplayer Rehabilitation Game Scored from Muscle and Brain, with Adaptive Score Manipulation.” Submitted to ACM CHI 2027.

Research Experience

RehabVerse: VR Motor Rehabilitation with Adaptive Scoring

Jun. 2026 – Sep. 2026

Research Assistant (Full-time), BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Investigated motor engagement during executed and imagined movements using mu-band event-related desynchronization (ERD) in 6 healthy participants.
- Developed stage-based analyses of EEG lateralization to assess responses following score boosts and led manuscript writing and figure preparation.

VR Motor Rehabilitation with Human and AI Competition

Jun. 2026 – Present

Research Assistant (Full-time), BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Conducted EEG experiments using the RehabVerse platform in a two-session study involving 22 healthy participants to compare solo, human-opponent, and AI-opponent conditions during executed and imagined movements.

Eye-Tracking and EEG-Based Communication Interface for ALS

Jun. 2026 – Present

Research Assistant (Full-time), BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Designed and fabricated a six-channel EEG headset for a planned ALS communication interface combining SSVEP-based selection and eye tracking.

Human-Centric Automated Driving: VR-Based Passenger Emotion Assessment

Sep. – Dec. 2025

Student Researcher (Part-time), BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Developed a Unity-based VR driving testbed to examine passenger valence and arousal across driving styles and road environments using EEG and self-reports.

Multimodal Control for an Assistive Mobile Manipulator

Jul. – Aug. 2025

Summer Research Intern, BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Designed and 3D-printed the mobile base of an EEG-controlled assistive robot that completed 2 object-retrieval tasks.
- Implemented MediaPipe-based control of an open-source 6-DOF arm, using gestures to select joints and lateral hand motion to control rotation.

Biomechanical Testing Platform for Small-Joint Evaluation

Jan. 2024 – Oct. 2025

Undergraduate Researcher, SAM Lab, NTU | Prof. Dian-Ru Li

- Designed a cadaveric limb motion platform for synchronized force measurement and optical tracking to assess mechanical changes under controlled joint defects.
- Used PCA and k-means clustering to group marker trajectories and quantified regional deformation through apparent in-plane strain analysis.

Interactive VR Simulator for Cardiac Catheterization Training

Jul. 2023 – Aug. 2024

Undergraduate Researcher, Advanced Medical Device Lab, NTU | Prof. Hao-Ming Hsiao

- Developed a Unity-based VR prototype for interactive guidewire positioning and balloon-driven stent expansion.

Additional Experience

Mechanical Engineering Intern, HP Inc.

Jul. 2024 – Jun. 2025

- Redesigned a laptop hinge to improve thermal performance and developed a modular SSD shielding cap adopted across 3 product lines, cutting replacement time by 86% compared with similar company designs.

Depth-Camera-Based Autonomous Driving (course project)

Feb. – Jun. 2025

- Trained a YOLOv8 object detector and developed a path-planning algorithm to support obstacle-aware navigation using a single depth camera.

Leadership Experience

Founder & President

Sep. 2024 – Jan. 2026

NTU Mechanical Engineering International Compass

- Founded a student organisation to expand access to overseas study information and led a team of up to 22 members that conducted 30+ alumni interviews and organised forums reaching 300+ attendees.

Honors and Awards

Dean’s List Award, NTU (top 5% of class)

Fall 2024, Spring 2025

Academic Excellence Award, NTHU (top 5% of class)

Fall 2021, Spring 2022

Technical Skills

Programming	Python, C# (Unity), MATLAB
ML & Vision	PyTorch, OpenCV, MediaPipe, YOLOv8
EEG Analysis	MNE-Python, EEGLAB
Hardware & VR	SolidWorks, Autodesk Inventor, 3D printing (FDM/SLA), Unity